

ABHISHEK AGARWAL

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PROFESSIONAL SUMMARY

Backend and GenAI Engineer with 3 years of experience building scalable backend systems, low-latency APIs, and AI-powered applications using FastAPI, LangGraph, and AWS. Experienced in production-grade RAG pipelines and distributed microservices.

EXPERIENCE

System Engineer, Tata Consultancy Services (TCS) Jan 2024 – Present

Enterprise Fraud Intelligence Platform (Agentic AI, Backend, Distributed Systems)

- Engineered scalable **FastAPI-based REST APIs** to automate transaction auditing, KYC verification, compliance validation, and fraud-risk analysis workflows for enterprise banking systems.
- Reduced fraud-investigation turnaround time by **45–55%** through automated risk analysis, policy validation, and intelligent workflow orchestration, improving operational efficiency and decision reliability.
- Orchestrated a **LangGraph-based agentic workflow** using **RAG**, transformer embeddings, and tool-driven execution for policy-aware fraud investigations across **MySQL** and **Redis** systems.
- Developed **Human-in-the-Loop (HITL)** escalation workflows for high-risk transactions, enabling secure manual review and reducing unsafe automated decision handling.
- Enhanced backend responsiveness using **Redis-based caching** and asynchronous processing, reducing repeated-query latency by **35–45%** while supporting scalable enterprise workloads.

Smart Transit Operations Platform (Agentic AI, Backend, Distributed Systems)

- Architected and delivered a **microservices-based backend system** exposing **REST APIs** and a GenAI-powered interface for real-time bus ETA, delay, and route-status queries.
- Optimized commuter wait-time visibility by **50–60%** through automated access to live operational data, improving service reliability and reducing dependency on manual coordination.
- Orchestrated an **agentic decision engine** using **RAG** and transformer embeddings to validate intents, enforce ID-level constraints, and execute tool-driven workflows involving **GPS**, **weather**, and **ETA** systems.
- Automated **proactive ETA-based notifications** using threshold-driven logic to trigger periodic alerts and alternate-route suggestions, improving commuter experience and operational responsiveness.
- Integrated **Redis-based caching** and **RabbitMQ** event pipelines to decouple synchronous APIs from background processing, improving system responsiveness and supporting scalable real-time operations.

PERSONAL PROJECTS

Enterprise Conversational Analytics Engine

- Built an LLM-powered conversational analytics engine enabling secure SQL query generation and natural language data exploration over relational databases.
- Implemented schema validation, query-safety enforcement, and role-based access control to prevent unauthorized database operations.

TECHNICAL SKILLS

Languages: Python, SQL, Core Java

Backend Engineering: REST APIs, API Development, FastAPI, Microservices, Distributed Systems, Redis

AI & MLOps: LangGraph, RAG, ChromaDB, Transformers, LangSmith, LLMs, CI/CD Pipelines, AWS

Cloud & Databases: MySQL, Vector Databases

ACHIEVEMENTS

- Created and published **Fundamentals of C** course on Udemy with **750+ global enrollments**.
- Achieved Global Rank **3915** (Weekly Contest 328) and **4026** (Biweekly Contest 103) on LeetCode.

EDUCATION

Bachelor of Technology in Information Technology

Academy of Technology, Kolkata

Aug 2019 – Jun 2023

CGPA: 9.37